

# CYBERSECURITY MASTERY ROADMAP

Beginner to Professional • SOC • Penetration Testing

■ Security+

■ CEH

■ OSCP

■ SIEM

■ Cloud Security

5

Learning Paths

100+

Tools Covered

50+

Certifications

300+

Resources

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# 01 — INTRODUCTION TO CYBERSECURITY

Cybersecurity is the practice of protecting systems, networks, programs, and data from digital attacks. In an era where data is the new oil, cybersecurity professionals are among the most sought-after experts globally. Whether you aim to defend organizations as a SOC Analyst, break into systems as an ethical hacker, or govern policies as a CISO, this roadmap provides a clear, structured path from complete beginner to professional.

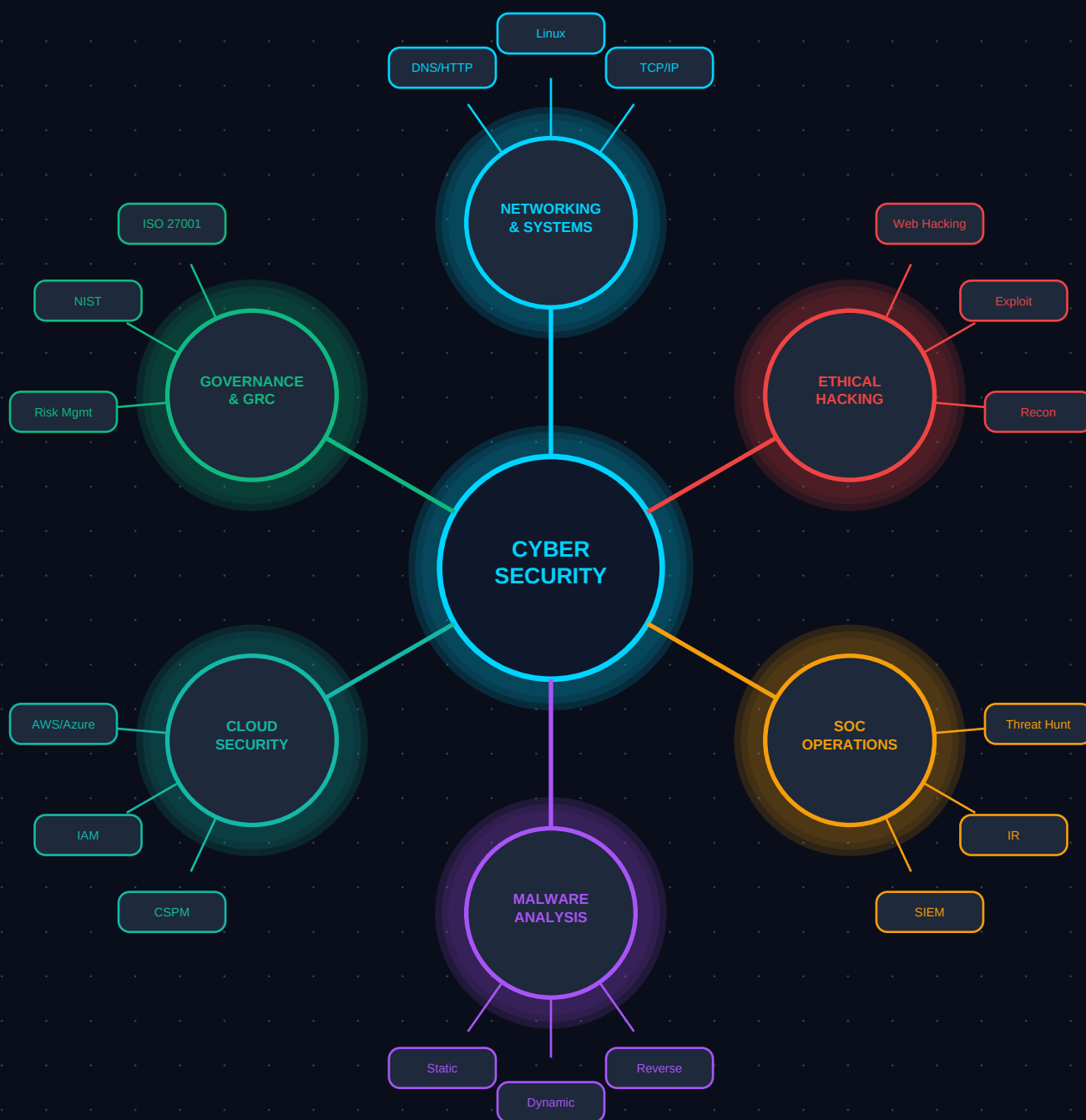
| Why Cybersecurity?                | Global Demand           | Salary Range (USD)             |
|-----------------------------------|-------------------------|--------------------------------|
| 3.5M+ unfilled positions globally | Growing 35% YoY         | \$65,000 – \$200,000+          |
| Every industry needs security     | Remote-friendly careers | Entry: \$65K   Mid: \$95K      |
| Constantly evolving field         | High job security       | Senior: \$130K   CISO: \$200K+ |

## ■ Core Cybersecurity Domains

|                        |                                                                     |
|------------------------|---------------------------------------------------------------------|
| ■ Network Security     | Protecting network infrastructure from unauthorized access, attacks |
| ■ Endpoint Security    | Securing individual devices like laptops, phones, servers           |
| ■ Application Security | Finding and fixing vulnerabilities in software and web apps         |
| ■ Cloud Security       | Securing cloud environments (AWS, Azure, GCP)                       |
| ■ Threat Intelligence  | Gathering and analyzing data about threats and threat actors        |
| ■ GRC                  | Governance, Risk & Compliance — policies, standards, auditing       |

## 02 — CYBERSECURITY MIND MAP

A holistic view of all interconnected cybersecurity domains — from your first day to expert level.

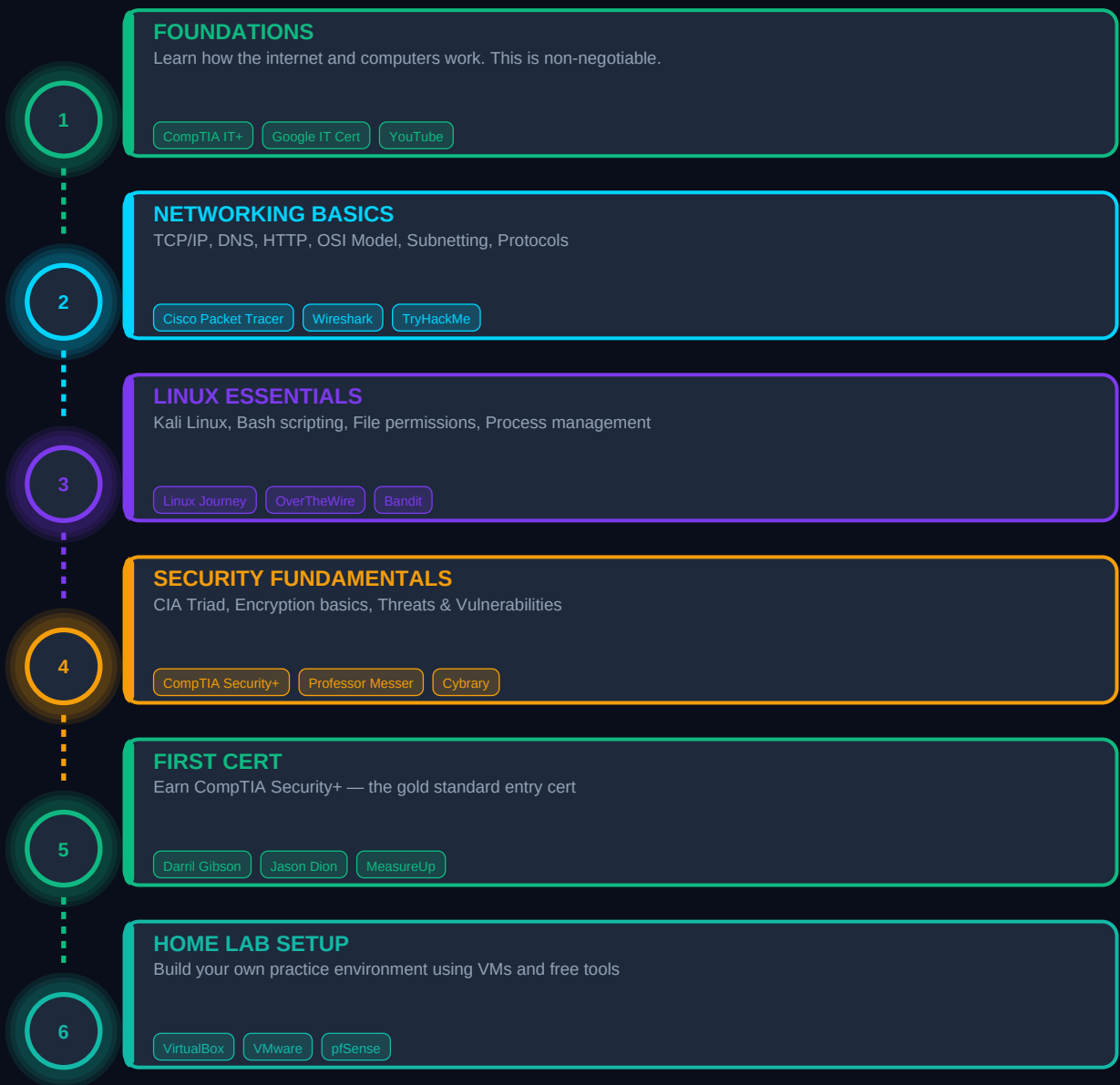


Cybersecurity Knowledge Mind Map — All domains interconnected

■ Every node in this map represents a critical skill area. Start from NETWORKING & SYSTEMS and expand outward as you progress.

# 03 — BEGINNER ROADMAP (0–6 Months)

Start here. No prior experience needed. Build your foundation strong.



## BEGINNER SKILL TARGETS (6 months)

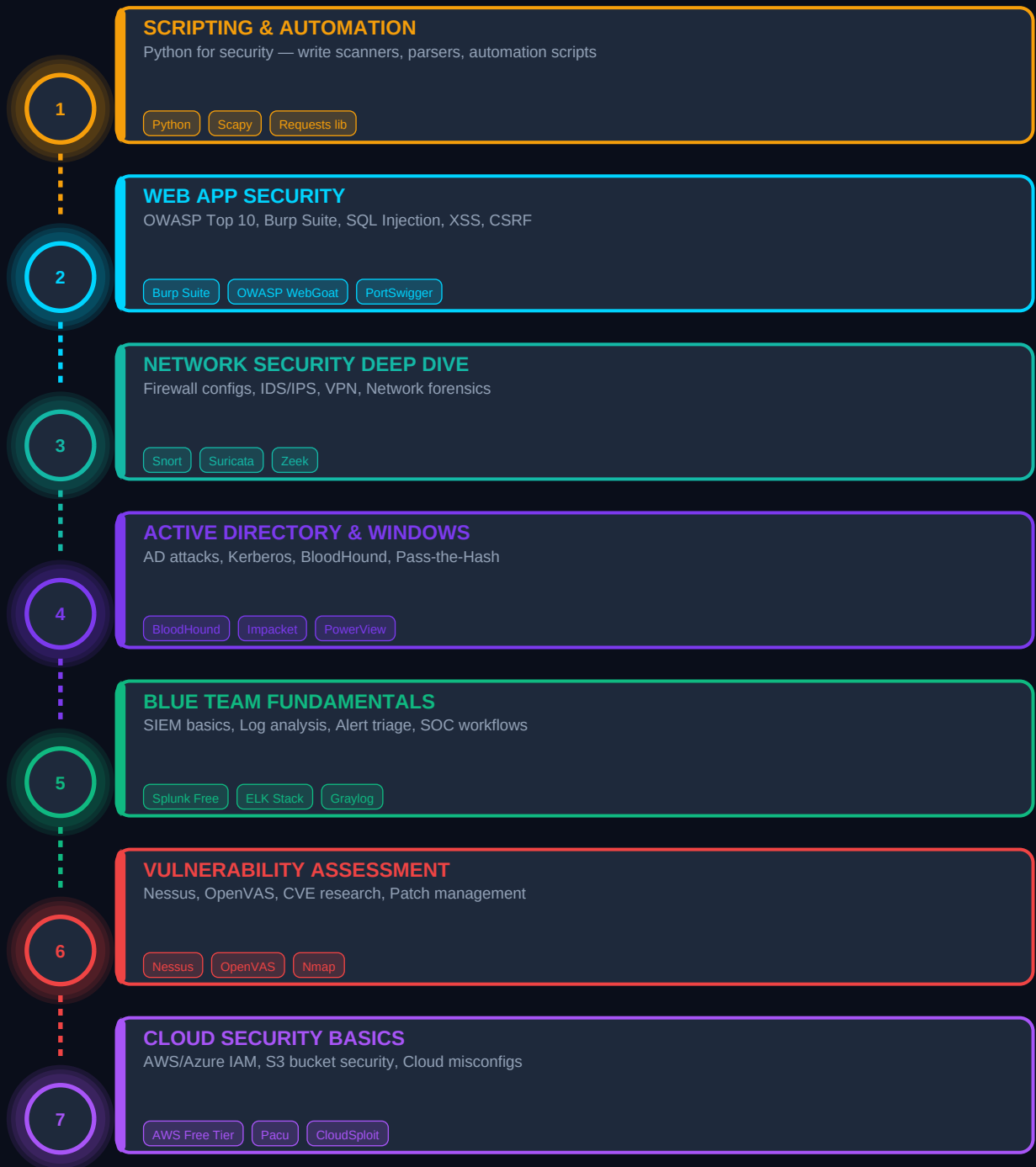
|                                                         |     |        |
|---------------------------------------------------------|-----|--------|
| <b>Networking</b><br>TCP/IP, OSI, Subnetting            | 40% | ●●●●●● |
| <b>Linux CLI</b><br>Bash, File System, Permissions      | 40% | ●●●●●● |
| <b>Security Concepts</b><br>CIA, Threats, Encryption    | 40% | ●●●●●● |
| <b>Python Basics</b><br>Variables, Loops, Functions     | 20% | ●●●●●● |
| <b>Windows Admin</b><br>AD basics, Registry, Event Logs | 20% | ●●●●●● |
| <b>Wireshark</b><br>Packet capture & analysis           | 20% | ●●●●●● |

## ■ Beginner Learning Resources

| Resource                         | Type             | Cost      | Focus                          |
|----------------------------------|------------------|-----------|--------------------------------|
| TryHackMe.com                    | Interactive Labs | Free/Paid | Beginner-friendly guided paths |
| Professor Messer (Security+)     | Video Course     | Free      | Security+ exam prep            |
| The Cyber Mentor (TCM)           | YouTube          | Free      | Ethical hacking fundamentals   |
| Hack The Box Academy             | Labs             | Free tier | Hands-on hacking practice      |
| Linux Journey (linuxjourney.com) | Tutorial         | Free      | Linux mastery                  |
| Cisco NetAcad                    | Course           | Free      | Networking fundamentals        |
| OverTheWire: Bandit              | Wargame          | Free      | Linux CLI challenges           |
| Cybrary.it                       | Video Course     | Free/Paid | Wide security curriculum       |
| SANS Cyber Aces                  | Tutorial         | Free      | OS & Networking basics         |

## 04 — INTERMEDIATE ROADMAP (6–18 Months)

Level up your skills. Start specializing. Build real projects.



## INTERMEDIATE SKILL TARGETS (18 months)

### Python Scripting

Security automation, parsers

60%

### Web App Hacking

OWASP Top 10, Burp Suite

60%

### Network Analysis

Wireshark, Zeek, Snort

60%

### Active Directory

AD attacks, Kerberos

40%

### SIEM/Log Analysis

Splunk, ELK, Graylog

60%

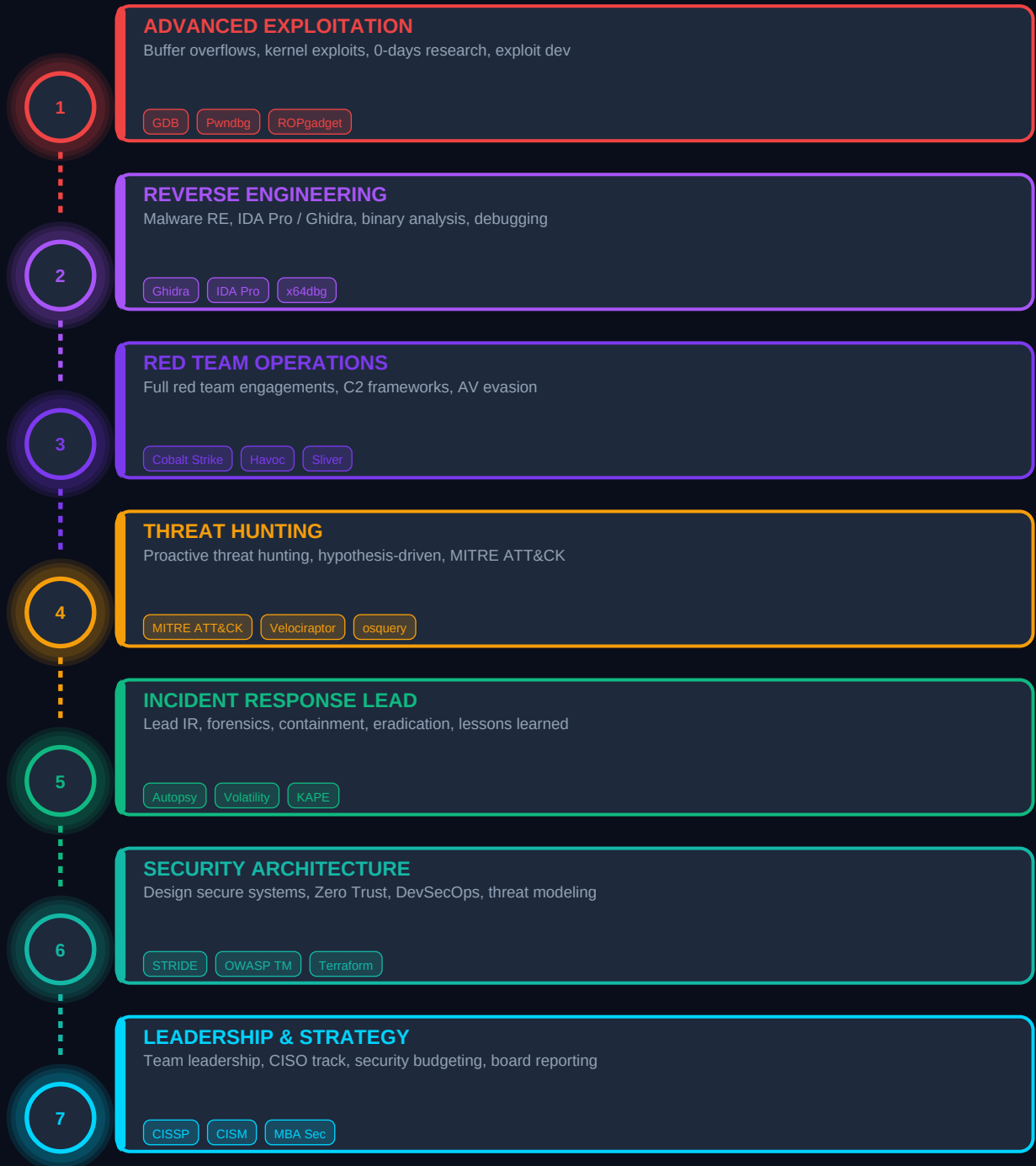
### Cloud Security

AWS IAM, S3, misconfigs

40%

## 05 — ADVANCED / PRO ROADMAP (18+ Months)

You are now dangerous (legally). Specialize, certify, lead teams.



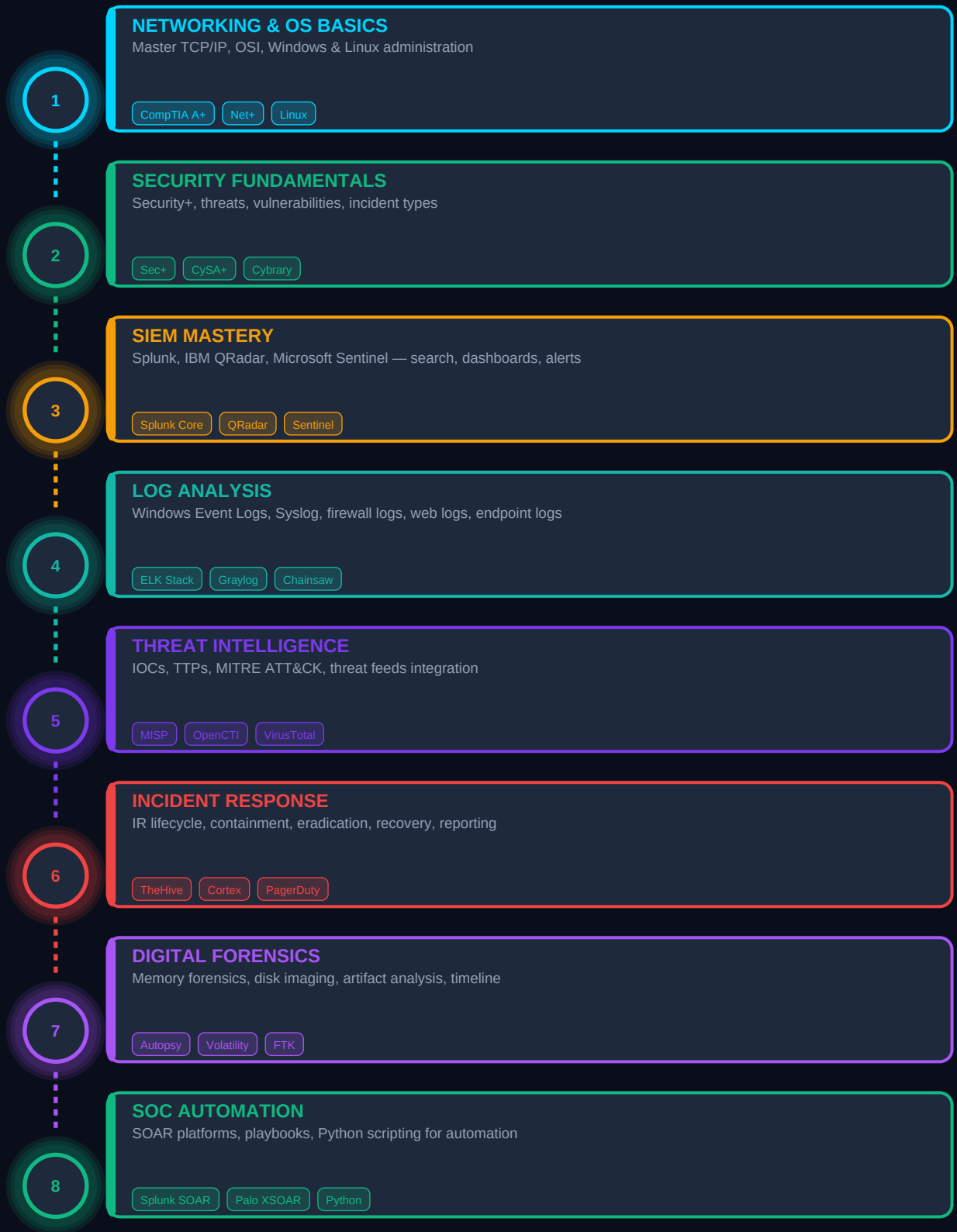


# 06 — SOC ANALYST CAREER PATH

A Security Operations Center (SOC) Analyst is the frontline defender. You monitor, detect, analyze, and respond to security incidents 24/7. This is one of the most in-demand entry roles in cybersecurity.

| SOC Tier               | Role             | Focus                                                | Avg Salary    |
|------------------------|------------------|------------------------------------------------------|---------------|
| Tier 1<br>(L1 Analyst) | Alert Triage     | Monitor SIEM alerts, basic investigation, escalation | \$55K–\$75K   |
| Tier 2<br>(L2 Analyst) | Incident Handler | Deep investigation, threat hunting, malware triage   | \$75K–\$100K  |
| Tier 3<br>(L3 Analyst) | Threat Hunter    | Proactive hunting, IR lead, tool development         | \$100K–\$140K |
| SOC Manager            | Team Lead        | Strategy, staffing, SLAs, executive reporting        | \$130K–\$180K |

## ■ SOC Analyst Roadmap — Step by Step



## ■ SOC Certifications Path



## ■ Essential SOC Tools

| Category     | Tools                                             | Purpose                                |
|--------------|---------------------------------------------------|----------------------------------------|
| SIEM         | Splunk, IBM QRadar, Microsoft Sentinel, ELK Stack | Log aggregation, correlation, alerting |
| Threat Intel | MISP, OpenCTI, VirusTotal, Shodan, OTX            | IOC enrichment, threat feeds           |
| Endpoint     | CrowdStrike, SentinelOne, Carbon Black, Defender  | EDR, malware detection                 |
| Network      | Zeek, Suricata, Snort, Arkime (Moloch)            | Network traffic analysis               |
| Forensics    | Autopsy, Volatility, FTK, Velociraptor, KAPE      | Memory & disk forensics                |
| IR/SOAR      | TheHive, Cortex, Palo Alto XSOAR, IBM SOAR        | Incident management & automation       |
| Ticketing    | ServiceNow, Jira, PagerDuty, Opsgenie             | Case management, alerting              |

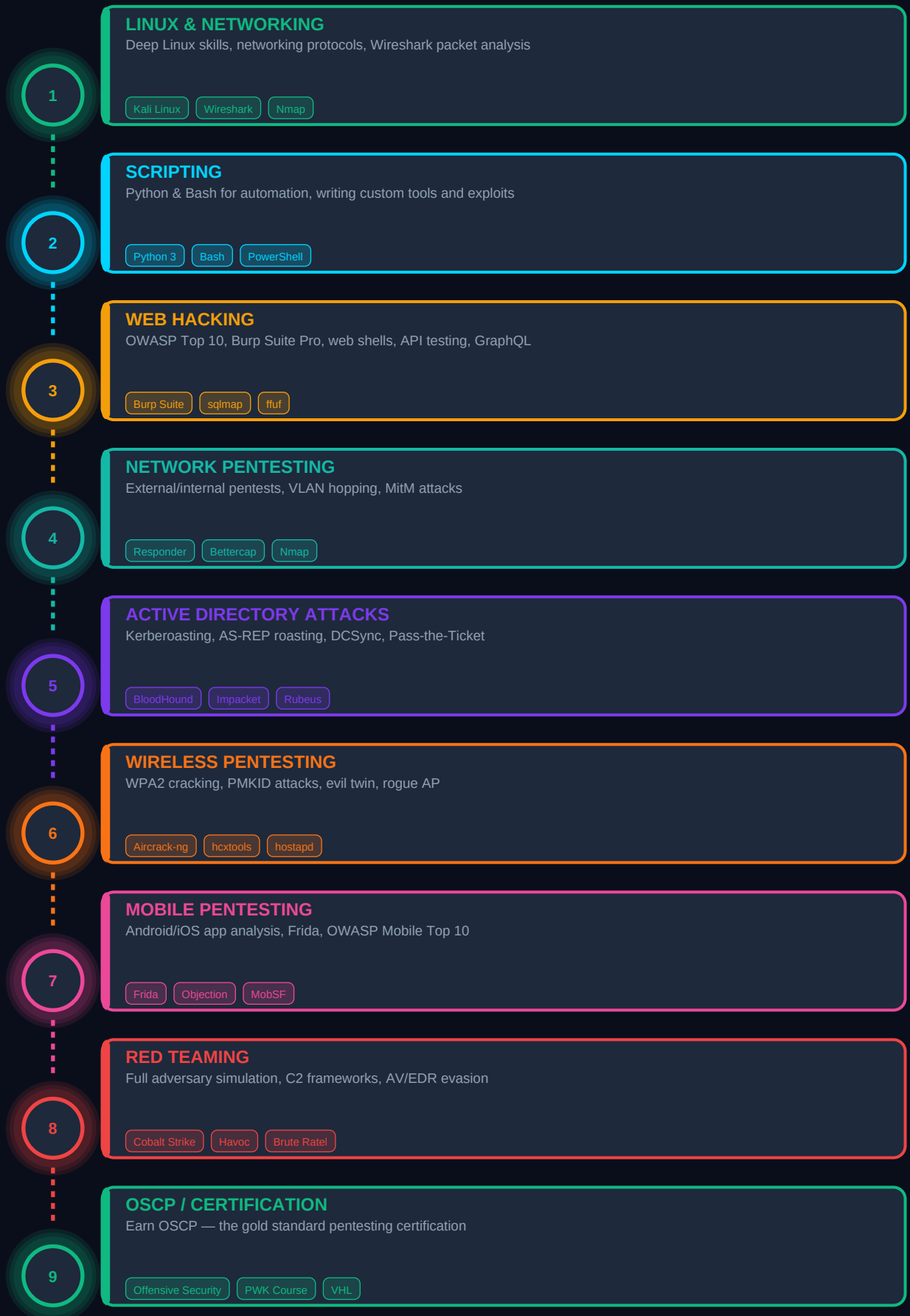
# 07 — PENETRATION TESTING CAREER PATH

Penetration Testing (ethical hacking) involves legally breaking into systems to find vulnerabilities before malicious actors do. This is one of the most exciting, creative, and well-paying careers in cybersecurity.

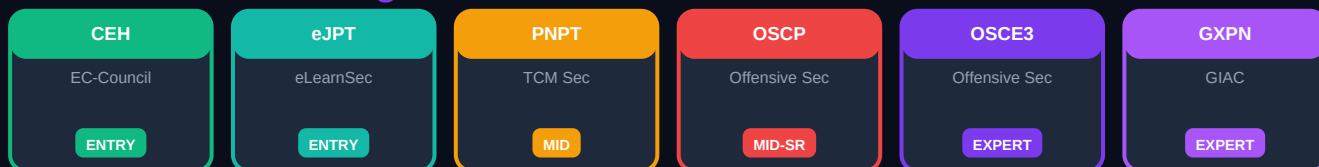
## ■ Pentest Methodology (PTES)

| Phase                     | Activities                                                    | Key Tools                                      |
|---------------------------|---------------------------------------------------------------|------------------------------------------------|
| 1. Pre-Engagement         | Scope definition, rules of engagement, legal agreements       | Statement of Work, NDA                         |
| 2. Reconnaissance         | OSINT, passive recon, DNS enum, employee profiling            | theHarvester, Maltego, Shodan, OSINT Framework |
| 3. Scanning & Enumeration | Port scanning, service detection, vulnerability scanning      | Nmap, Nessus, Masscan, Netcat                  |
| 4. Exploitation           | Exploit vulnerabilities, gain initial access, bypass defenses | Metasploit, Exploit-DB, custom exploits        |
| 5. Post-Exploitation      | Privilege escalation, lateral movement, persistence           | Mimikatz, BloodHound, PowerShell Empire        |
| 6. Reporting              | Document findings, risk ratings, remediation guidance         | Dradis, Plextrac, LaTeX reports                |

## ■ Penetration Testing Roadmap



## ■ Penetration Testing Certifications



## ■ MITRE ATT&CK; Tactics — Attacker Mindset

| Tactic               | What attackers do                              | Defense                                    |
|----------------------|------------------------------------------------|--------------------------------------------|
| Reconnaissance       | OSINT, scanning, social engineering            | Reduce attack surface, monitor exposure    |
| Resource Development | Build/buy infra, acquire accounts, malware     | Threat intel, dark web monitoring          |
| Initial Access       | Phishing, exploit public apps, supply chain    | MFA, patch management, email filtering     |
| Execution            | Scripts, WMI, scheduled tasks, living-off-land | App whitelisting, EDR, behavioral analysis |
| Persistence          | Registry keys, startup items, backdoors        | Integrity monitoring, EDR, user auditing   |
| Privilege Escalation | Exploit vulns, token impersonation             | Least privilege, patch management          |
| Defense Evasion      | Obfuscation, disable tools, LOLBAS             | EDR, honeypots, behavioral analytics       |
| Credential Access    | Mimikatz, keylogging, credential dumping       | Credential Guard, MFA, PAM                 |
| Lateral Movement     | Pass-the-Hash, RDP, WMI, BloodHound paths      | Network segmentation, Zero Trust           |
| Exfiltration         | DNS tunneling, HTTPS, cloud storage abuse      | DLP, traffic analysis, CASB                |

## 08 — COMPLETE CERTIFICATIONS GUIDE

| Certification          | Vendor             | Level        | Domain                | Cost (USD) | Validity |
|------------------------|--------------------|--------------|-----------------------|------------|----------|
| CompTIA A+             | CompTIA            | Beginner     | IT Fundamentals       | \$250      | 3 yrs    |
| CompTIA Network+       | CompTIA            | Beginner     | Networking            | \$350      | 3 yrs    |
| CompTIA Security+      | CompTIA            | Beginner     | Security Fundamentals | \$392      | 3 yrs    |
| CompTIA CySA+          | CompTIA            | Intermediate | SOC/Blue Team         | \$392      | 3 yrs    |
| CompTIA PenTest+       | CompTIA            | Intermediate | Pentesting            | \$392      | 3 yrs    |
| CompTIA CASP+          | CompTIA            | Advanced     | Security Architecture | \$494      | 3 yrs    |
| CEH                    | EC-Council         | Intermediate | Ethical Hacking       | \$1,200    | 3 yrs    |
| OSCP                   | Offensive Security | Advanced     | Pentesting            | \$1,499    | Lifetime |
| OSCE3                  | Offensive Security | Expert       | Advanced Exploitation | \$1,649    | Lifetime |
| CISSP                  | ISC2               | Expert       | Security Management   | \$749      | 3 yrs    |
| CISM                   | ISACA              | Expert       | Security Management   | \$575      | 3 yrs    |
| GCIH                   | GIAC               | Intermediate | Incident Response     | \$949      | 4 yrs    |
| GPEN                   | GIAC               | Advanced     | Pentesting            | \$949      | 4 yrs    |
| GCFA                   | GIAC               | Advanced     | Forensics             | \$949      | 4 yrs    |
| eJPT                   | eLearnSecurity     | Beginner     | Pentesting            | \$200      | 2 yrs    |
| PNPT                   | TCM Security       | Intermediate | Pentesting            | \$400      | 2 yrs    |
| AWS Security Specialty | Amazon             | Advanced     | Cloud Security        | \$300      | 3 yrs    |
| AZ-500                 | Microsoft          | Intermediate | Azure Security        | \$165      | Lifetime |
| BTL1                   | Security Blue Team | Intermediate | Blue Team/SOC         | \$540      | Lifetime |
| Splunk Core Certified  | Splunk             | Intermediate | SIEM/SOC              | \$130      | 3 yrs    |

## 09 — TOOLS ARSENAL

### ■ Reconnaissance

| Tool         | Description                                  |
|--------------|----------------------------------------------|
| Nmap         | Network scanner — king of port scanners      |
| Shodan       | Search engine for internet-connected devices |
| theHarvester | Email, domain, IP OSINT gathering            |
| Maltego      | Visual link analysis and OSINT               |
| Recon-ng     | Web reconnaissance framework                 |
| Amass        | Subdomain enumeration and network mapping    |
| SpiderFoot   | Automated OSINT intelligence gathering       |

### ■ Exploitation

| Tool                      | Description                               |
|---------------------------|-------------------------------------------|
| Metasploit Framework      | World's most used exploitation framework  |
| Exploit-DB / SearchSploit | Vulnerability and exploit database        |
| Burp Suite Pro            | Web application security testing platform |
| SQLmap                    | Automated SQL injection and takeover tool |
| BeEF                      | Browser Exploitation Framework            |
| Responder                 | LLMNR/NBT-NS/MDNS poisoner for creds      |
| CrackMapExec              | Swiss army knife for network pentesting   |

### ■ Defense & Monitoring

| Tool             | Description                                 |
|------------------|---------------------------------------------|
| Splunk           | Industry-leading SIEM platform              |
| ELK Stack        | Open-source log management (Elastic)        |
| Snort / Suricata | Open-source IDS/IPS engines                 |
| Zeek (Bro)       | Network analysis framework                  |
| Wazuh            | Open-source security platform / SIEM        |
| OSSEC            | Open-source HIDS — host intrusion detection |
| Graylog          | Open-source log management                  |

### ■ Forensics & Malware

| Tool    | Description                              |
|---------|------------------------------------------|
| Autopsy | Digital forensics platform (open source) |

|                    |                                        |
|--------------------|----------------------------------------|
| Volatility 3       | Memory forensics framework             |
| Ghidra             | NSA's reverse engineering tool (free!) |
| IDA Pro / Ida Free | Industry-standard disassembler         |
| KAPE               | Kroll Artifact Parser and Extractor    |
| FTK Imager         | Forensic disk imaging tool             |
| Cuckoo Sandbox     | Automated malware analysis             |

## 10 — LEARNING RESOURCES & LABS

| Platform                | URL                                                                             | Type                   | Level           | Cost      |
|-------------------------|---------------------------------------------------------------------------------|------------------------|-----------------|-----------|
| TryHackMe               | <a href="https://tryhackme.com">tryhackme.com</a>                               | Guided Labs            | Beginner–Mid    | Free/Paid |
| Hack The Box            | <a href="https://hackthebox.com">hackthebox.com</a>                             | Challenge Labs         | Mid–Expert      | Free/Paid |
| VulnHub                 | <a href="https://vulnhub.com">vulnhub.com</a>                                   | Offline VMs            | All levels      | Free      |
| PortSwigger Web Academy | <a href="https://portswigger.net/web-security">portswigger.net/web-security</a> | Web Hacking            | Beginner–Expert | Free      |
| PentesterLab            | <a href="https://pentesterlab.com">pentesterlab.com</a>                         | Web & Code             | Beginner–Expert | Free/Paid |
| SANS Cyber Ranges       | <a href="https://sans.org/cyberaces">sans.org/cyberaces</a>                     | NetWars, etc.          | Mid–Expert      | Paid      |
| TCM Security Academy    | <a href="https://academy.tcm-sec.com">academy.tcm-sec.com</a>                   | Video + Labs           | Beginner–Mid    | Paid      |
| Offensive Security PG   | <a href="https://offensive-security.com">offensive-security.com</a>             | Proving Grounds        | Mid–Expert      | Paid      |
| Blue Team Labs Online   | <a href="https://blueteamlabs.online">blueteamlabs.online</a>                   | SOC/Defensive          | Beginner–Mid    | Free/Paid |
| CyberDefenders          | <a href="https://cyberdefenders.org">cyberdefenders.org</a>                     | Blue Team CTFs         | Mid–Expert      | Free      |
| LetsDefend.io           | <a href="https://letsdefend.io">letsdefend.io</a>                               | SOC Analyst Sim        | Beginner–Mid    | Free/Paid |
| Root-Me                 | <a href="https://root-me.org">root-me.org</a>                                   | Challenges             | All levels      | Free      |
| PicoCTF                 | <a href="https://picoctf.org">picoctf.org</a>                                   | CTF (beginner)         | Beginner        | Free      |
| OWASP WebGoat           | <a href="https://owasp.org/WebGoat">owasp.org/WebGoat</a>                       | Intentionally Vuln App | Beginner        | Free      |
| DVWA                    | <a href="https://dvwa.co.uk">dvwa.co.uk</a>                                     | Intentionally Vuln App | Beginner        | Free      |
| Metasploitable          | <a href="https://sourceforge.net">sourceforge.net</a>                           | Vulnerable VM          | Beginner–Mid    | Free      |

### ■ Essential YouTube Channels

| Channel                     | Focus                                                   |
|-----------------------------|---------------------------------------------------------|
| The Cyber Mentor (TCM)      | Ethical hacking, OSCP prep, pentesting methodology      |
| NetworkChuck                | Networking, Linux, cloud — beginner-friendly & engaging |
| John Hammond                | CTF walkthroughs, malware analysis, TryHackMe           |
| IppSec                      | Hack The Box machine walkthroughs — post-retirement     |
| David Bombal                | Networking, Python, ethical hacking                     |
| LiveOverflow                | Binary exploitation, CTFs, deep security topics         |
| STÖK                        | Bug bounty hunting, web hacking, community              |
| 13Cubed                     | DFIR, Windows forensics, memory forensics               |
| Gerald Auger (Simply Cyber) | SOC, GRC, career advice, daily news                     |
| Professor Messer            | CompTIA certification prep (free videos)                |

# 11 — GITHUB PORTFOLIO STRATEGY

A strong GitHub portfolio is your resume in cybersecurity. It shows employers you don't just know concepts — you build things, document your work, and share knowledge.

| Project Idea                  | Skills Demonstrated                      | Difficulty   |
|-------------------------------|------------------------------------------|--------------|
| Network Scanner (Python)      | Nmap wrapper, socket programming, Python | Beginner     |
| Password Strength Checker     | Python, regex, security awareness        | Beginner     |
| Web Vulnerability Scanner     | HTTP requests, OWASP, Python             | Intermediate |
| Log Parser & Alert System     | Regex, Python, SIEM concepts             | Intermediate |
| Keylogger (educational)       | Python, OS interaction, malware concepts | Intermediate |
| CTF Writeups Repo             | Documentation, problem-solving, tools    | Any Level    |
| TryHackMe Progress Tracker    | Markdown docs, consistency               | Beginner     |
| OSINT Automation Tool         | APIs, Python, web scraping               | Intermediate |
| Threat Intelligence Dashboard | Python, Dash/Streamlit, APIs             | Advanced     |
| Custom C2 Framework (basic)   | Networking, Python, red team concepts    | Advanced     |
| Malware Analysis Lab Setup    | VM config, FLARE VM, documentation       | Advanced     |
| Home SOC Lab with ELK Stack   | Docker, ELK, networking, syslog          | Advanced     |
| Bug Bounty Findings Report    | Web hacking, professional reporting      | Expert       |

## ■ GitHub Best Practices for Security Professionals

- Pin 6 of your best repositories to your profile
- Write detailed README.md files with screenshots and setup instructions
- Add topics/tags to all repos (e.g., cybersecurity, ethical-hacking, python)
- Document your CTF writeups — they show problem-solving process
- NEVER commit real credentials, API keys, or sensitive targets to public repos
- Contribute to open-source security tools (even docs/testing)
- Star and fork tools you use — it shows your toolset
- Keep a consistent commit history — consistency matters to employers
- Create a personal cybersecurity notes repo (like a public notebook)
- Add a professional profile README with your goals and learning path

## 12 — CTF COMPETITIONS & PRACTICE LABS

Capture The Flag (CTF) competitions are the best way to practice real hacking skills in a legal, gamified environment. They build problem-solving, persistence, and creativity.

| CTF Category              | Description                                                 | Skills Needed                  |
|---------------------------|-------------------------------------------------------------|--------------------------------|
| Web Exploitation          | Find bugs in web applications — SQLi, XSS, SSRF, RCE        | Burp Suite, HTML, JS, PHP      |
| Binary Exploitation (pwn) | Buffer overflows, format strings, ROP chains, heap exploits | C, GDB, Python, pwntools       |
| Reverse Engineering       | Disassemble and understand compiled binaries, firmware      | Ghidra, IDA, x64dbg, assembly  |
| Cryptography              | Break weak crypto, implement attacks on RSA, AES, hashes    | Math, Python, crypto theory    |
| Forensics                 | Analyze images, memory dumps, network captures              | Wireshark, Autopsy, Volatility |
| OSINT                     | Find information using only public sources                  | Google dorks, Shodan, Maltego  |
| Steganography             | Find hidden data in images, audio, files                    | steghide, binwalk, strings     |
| Miscellaneous             | Programming challenges, trivia, unique puzzles              | Varies                         |

### ■ Top CTF Competitions to Enter

| Competition           | Organizer                  | Level                 | When            |
|-----------------------|----------------------------|-----------------------|-----------------|
| picoCTF               | Carnegie Mellon University | Beginner              | Annual (Spring) |
| CTFtime.org events    | Community                  | All levels            | Year-round      |
| DEFCON CTF            | DEFCON Conference          | Expert                | Annual (August) |
| SANS Holiday Hack     | SANS Institute             | All levels            | December        |
| Google CTF            | Google Security            | Advanced              | Annual          |
| Hack The Box CTF      | HackTheBox                 | Intermediate–Advanced | Multiple/year   |
| NahamCon CTF          | NahamSec                   | Beginner–Mid          | Annual          |
| BSides Events         | Various Cities             | All levels            | Year-round      |
| National Cyber League | NCL                        | Beginner–Mid          | Biannual (US)   |

## ■ FINAL WORDS

Cybersecurity is a marathon, not a sprint. The field rewards curiosity, persistence, and continuous learning. Every expert was once a beginner who refused to give up. Start with the fundamentals, build your lab, join communities (Discord, Reddit *r/netsec*, Twitter/X), participate in CTFs, document everything on GitHub, and keep building. The hacking community is welcoming to beginners who show genuine effort and ethical intent. Remember: with great power comes great responsibility. Always hack legally and ethically.

|                 |                                                          |
|-----------------|----------------------------------------------------------|
| <b>Week 1–2</b> | Set up Kali Linux VM, complete Linux basics on TryHackMe |
| <b>Week 3–4</b> | Networking fundamentals — TCP/IP, OSI, Wireshark         |
| <b>Month 2</b>  | Start Security+ study — Professor Messer videos          |
| <b>Month 3</b>  | Complete TryHackMe Pre-Security path (free)              |
| <b>Month 4</b>  | Start Hack The Box Starting Point machines               |
| <b>Month 5</b>  | Pick SOC OR Pentest track — start specialization         |
| <b>Month 6</b>  | Book your first certification exam                       |
| <b>Month 7+</b> | Enter your first CTF, start building GitHub projects     |